INSURE

Consented Real-World IoT Telematics Insurance Pricing Platform

Actuarial Problem Alignment, Mathematical Proofs & Judge Evaluation Guide

INSURE Engineering & Actuarial Team • Published October 2026 • Production Certified

ABSTRACT & EXECUTIVE SUMMARY

Traditional automobile insurance pricing relies on demographic proxies (age, gender, postal code, credit history) that penalize safe drivers and fail to capture actual driving risk. We present **INSURE**, an actuarially calibrated, privacy-preserving IoT telematics platform. INSURE ingests real-world smartphone and OBD-II vehicle telemetry with uncoerced consent, defensively validates noisy sensor records, calculates deterministic risk scores across 5 physical driving factors, computes transparent quotes with itemized savings, and actively audits demographic fairness. Under the Section 8 growth benchmark, this model lowers policyholder premiums by 10% while increasing net insurer underwriting profit by **+63.33%** through adverse selection elimination.

1. Problem Statement & Challenge Alignment

Problem & Challenge Area	Traditional Industry Flaw	INSURE Engineered Solution
Consented Real-World Behavioral Data	Crude demographic proxies (age, postal code) fail to reflect true on-road risk.	Ingests real-world vehicle telematics (OBD-II, smartphone IMU, GPS) strictly with explicit user consent controls.
Noisy or Incomplete IoT Data	GPS multi-path drift, erratic sampling, and sub-kilometer test trips create false risk spikes.	**Defensive Ingestion Validator**: filters GPS jitter, verifies monotonic timestamps ($\Delta t > 0$), and requires minimum exposure (≥ 5 km, ≥ 15 min) before scoring.
Privacy Concerns & Monitoring	Drivers fear constant surveillance and secondary monetization of raw route data.	**Granular Consent & Data Minimization**: Policyholders can toggle GPS resolution or delete traces. System stores aggregated metrics, discarding continuous raw breadcrumbs.
Unexplained Premium Fluctuations	Opaque black-box ML models output quotes with zero explainability, eroding customer trust.	**Deterministic Factor Attribution**: Deconstructs every quote into 5 explicit physical driving factors. Every rupee saved or charged is itemized with plain-language rationale.
Model Fairness & Discrimination	Unsupervised models inadvertently penalize night-shift workers or socioeconomic groups.	**Fairness Audit Engine**: Continuous disparate impact ratio monitoring (80% Four-Fifths Rule compliance). Legitimate accident claims are strictly decoupled from scoring.
Human Review & Dispute Channel	Emergency evasive maneuvers or sensor faults unfairly penalize safe policyholders.	**Human-in-the-Loop Dispute Workflow**: Policyholders can flag any trip in 1 click; underwriters review raw sensor logs, adjust scores, and issue automated premium refunds.

2. Complete Mathematical Model with Real Sample Data

Step 1: Defensive Data Quality Ingestion Validation

A driver completes an urban commute generating raw vehicle telemetry. The defensive validator runs:

- **Trip Exposure:** Distance = 14.2 km • Duration = 22 minutes (1,320 seconds) • Status: ≥ 5 km threshold met.
- **GPS Fix Quality:** 98.4% valid coordinates ($\Delta t > 0$ strictly monotonic). Quality score $Q = 0.984 \geq 0.85 \rightarrow$ **ACCEPTED**.
- **Detected Sensor Events:** Harsh Braking ($\leq$ -3.8 m/s²): 1 event • Rapid Acceleration ($\geq$ +3.5 m/s²): 2 events • Speeding (>15 km/h over posted limit): 0.8 km (5.6% of trip) • Late-Night Driving (23:00 - 05:00): 0.0 km (0%) • In-Motion Phone Handling: 0.2 minutes (1.5% of duration).

Step 2: 5-Factor Actuarial Risk Scoring Formula

```
Factor Score_i = max(0, 100 − Deductions_i)  |  Overall Safety Score S = ∑ (w_i × Factor Score_i)
```

Factor Name	Weight (w)	Raw Event Metric	Deduction Formula	Factor Score	Weighted Score
1. Harsh Braking	0.25	1 event in 14.2 km (0.07/km)	Deduct 10 × (events / 10 km) = -0.7	93.00	**23.25**
2. Rapid Acceleration	0.20	2 events in 14.2 km (0.14/km)	Deduct 8 × (events / 10 km) = -1.1	88.87	**17.78**
3. Excessive Speeding	0.25	5.6% distance over limit	Deduct 1.5 × (% speeding) = -8.4	91.60	**22.90**
4. Late-Night Exposure	0.15	0.0% night driving	Deduct 1.0 × (% night) = 0.0	100.00	**15.00**
5. Phone Distraction	0.15	1.5% trip time handling phone	Deduct 2.0 × (% distraction) = -3.0	97.00	**14.55**
Total	**1.00**	**Overall Actuarial Safety Score: 93.48 / 100 (Rounds to 93)**			**Tier: LOW RISK**

Step 3: Transparent Dynamic Premium Calculation

Approved Insurer Baseline Annual Commercial Premium: **P_base = ₹2,000**.

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Actuarial Discount Schedule:
• Safety Score ≥ 90: 10% to 15% Safe Driver Discount (Applied: 10% for Score = 93)
• 75 ≤ Safety Score < 90: 5% to 10% Moderate Discount
• 60 ≤ Safety Score < 75: 0% Standard Market Rate
• Safety Score < 60: Regulatory Capped Surcharge (≤ +20%)

Final Annual Premium = P_base × (1 − Discount Rate) = ₹2,000 × (1 − 0.10) = ₹1,800
Net Policyholder Annual Savings = ₹2,000 − ₹1,800 = ₹200 / year (10% Savings)
```

Itemized Policyholder Statement (Presented in Portal)

- Approved Commercial Base Premium: ₹2,000
- Zero Late-Night Exposure Discount: − ₹80
- High Speed-Limit Compliance Discount: − ₹70
- Low Distraction Phone Usage Discount: − ₹50
- Braking and Acceleration Variance: + ₹0
- **Final Transparent Annual Quote: ₹1,800**

Step 4: Actuarial Portfolio Economics (Section 8 Benchmark Proof)

The Underwriting Paradox Solved: How does reducing average premium from ₹2,000 to ₹1,800 create **+63.33% higher profit**?

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Underwriting Profit = Premium Revenue − Claims Incurred − Total Expenses
Combined Ratio = (Claims Incurred + Total Expenses) / Premium Revenue × 100%
```

Financial Metric	Before INSURE (Baseline)	With INSURE (Section 8)	Mathematical Explanation
1. Policyholders (N)	1,000 drivers	**1,300 drivers (+30%)**	Transparent pricing attracts safe drivers and boosts retention.
2. Average Premium (P)	₹2,000	**₹1,800 (-10%)**	Competitive safe-driver rate.
3. Premium Revenue (N × P)	₹20,00,000	**₹23,40,000 (+₹3,40,000)**	Volume growth outpaces the 10% discount (1.3 × 0.9 = 1.17).
4. Claims Incurred (Losses)	₹12,00,000 (60.0% claims ratio)	**₹13,00,000 (55.55% claims ratio)**	Telematics feedback reduces crash frequency from 6.0% to 4.2%.
5. Operating + Platform + CAC	₹5,00,000 (25.0% expense ratio)	**₹5,50,000 (23.50% expense ratio)**	Cloud automation yields operational scale efficiencies.
6. Total Expenses	₹17,00,000	**₹18,50,000**	Combined claims incurred and operating expenses.
7. Combined Ratio	**85.00%**	**79.06% (-5.94%)**	Combined Ratio < 100% generates healthy underwriting margin.
8. Net Underwriting Profit	**₹3,00,000 (15.0% margin)**	**₹4,90,000 (20.94% margin)**	**+₹1,90,000 (+63.33% Net Profit Surge!)**

3. Role-by-Role Demonstration Flow for Judges

ROLE 1: CUSTOMER **tharunkumark4743@gmail.com (1-Click Login)**

- **Privacy & Consent (/consent):** Highlights uncoerced opt-in, toggleable GPS resolution, and complete data minimization.
- **IoT Driving Simulator (/simulator):** Runs live 5-stage telematics ingestion, quality validation, 5-factor scoring, and accident separation.
- **Transparent Billing & Dispute (/assessments):** Displays itemized factor deductions and triggers 1-click human dispute dispatch.

ROLE 2: ADMIN / UNDERWRITER **tharunkumark42007@gmail.com (1-Click Login)**

- **Portfolio Economics (/profitability):** Interactive sandbox comparing Baseline, Section 8, and Optimistic models with live ratio recalculation.
- **Pricing Optimizer:** Multi-scenario evaluator finding optimal regulatory-capped profit strategies.
- **Underwriting Review Panel (/review):** Underwriters inspect raw event logs, override unfair penalties, and issue automated refunds.

ROLE 3: COMPLIANCE OFFICER **vsbecitc2428@gmail.com (1-Click Login)**

- **Fairness & Parity Audit (/audit):** Verifies disparate impact ratios (80% Four-Fifths compliance) ensuring zero demographic discrimination.
- **Export Custom Report:** Instant export of signed CSV and JSON compliance audit packs for regulatory inspection.

4. Commercial Feasibility, Viability & Strategic Moat

- **Hardware Agnostic:** Compatible with standard smartphone accelerometer/gyroscope/GPS or plug-and-play OBD-II dongles.
- **Serverless Architecture:** High-throughput Node.js/TypeScript and React/Vite architecture running with virtually zero standby cost.
- **Aligned Incentives:** Safe drivers save up to 15%, while insurers earn +63.33% profit via risk self-selection.
- **Legitimate Claims Protection:** Behavioral scores only discount premiums; legitimate accident claims are never delayed or denied.